

Kinases and kinase signaling pathways: Potential therapeutic targets in Parkinson's disease

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ABSTRACT

Complex molecular mechanisms underlying the pathogenesis of Parkinson's disease (PD) are gradually being elucidated. Accumulating genetic evidence implicates dysfunction of kinase activities and phosphorylation pathways in the pathogenesis of PD. Causative and risk gene products associated with PD include protein kinases (such as PINK1, LRRK2 and GAK) and proteins related phosphorylation signaling pathways (such as SNCA, DJ-1). PINK1, LRRK2 and several PD gene products have been associated with mitogen-activated protein (MAP) and protein kinase B (AKT) kinase signaling pathways. C-Jun N-terminal kinase (JNK), extracellular signal-regulated kinases (ERK) and p38, signaling pathways downstream of MAP, are particularly important in PD. JNK and p38 play an integral role in neuronal death. Targeting JNK or p38 signaling may offer an effective therapy for PD. Inhibitors of the ERK signaling pathway, which plays an important role in the development of L-DOPA-induced dyskinesia (LID), have been shown to attenuate this condition in animal models. In this review, we summarize experimental evidence gathered over the last decade on the role of PINK1, LRRK2 and GAK and their related phosphorylation signaling pathways (JNK, ERK, p38 and PI3K/AKT) in PD. It is speculated that improvement or modulation of these signaling pathways will reveal potential therapeutic targets for attenuation of the cardinal symptoms and motor complications in patients with PD in the future.

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Abbreviations: AD, autosomal dominant; ALP, autophagy-lysosome pathway; AR, autosomal recessive; CCVs, clathrin-coated vesicles; CDK, cyclin-dependent kinase; DA, dopamine; ERK, extracellular signal-regulated kinases; 4E-BP, 4E (eIF4E)-binding protein; GAK, cyclin-G-associated kinase; GSK, glycogen synthase kinase; IGF-1, insulin-like growth factor-I; JIPs, JNK-interacting proteins; JNK, C-Jun N-terminal kinase; LID, levodopa induced dyskinesia; LRRK2, leucine-rich repeat kinase 2; MAP, mitogen-activated protein; MPP⁺, 1-methyl-4-phenylpyridinium; MPTP, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine; 6-OHDA, 6-hydroxydopamine; PINK1, PTEN-induced kinase 1; PD, Parkinson's disease; PTPs, permeability transition pores; SNpc, substantia nigra pars compacta; TH, tyrosine hydroxylase; TRAP1, tumor necrosis factor receptor-associated protein-1; UCH-L1, ubiquitin-C-terminal hydrolase-L1; UPS, ubiquitin-proteasome system.

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1. Introduction

Parkinson's disease (PD) is a common neurodegenerative disorder characterized by selective loss of dopaminergic neurons in the substantia nigra and formation of intracellular inclusions known as Lewy bodies in the remaining cells (Braak et al., 2004; Greenamyre and Hastings, 2004). Through identification of the genes involved in familial and sporadic PD during the past ten years, the pathogenesis of PD has been related to impairments of the ubiquitin–proteasome system (UPS) (Cook and Petrucci, 2009; Matsuda and Tanaka, 2010), the autophagy-lysosome pathway (ALP) (Pan et al., 2008), mitochondrial metabolism disturbance (Cali et al., 2011; Ho et al., 2012) and alternated kinase activity (Cookson et al., 2007). Among these systems, the roles of kinases and kinase pathways have become an increasing focus of research and mutations in several genes encoding protein kinases and proteins involved in phosphorylation signaling pathways have been unequivocally linked to PD. Such protein kinases include PTEN-induced kinase 1 (PINK1) (Valente et al., 2001, 2004), leucine-rich repeat kinase 2 (LRRK2) (Funayama et al., 2002; Kumari and Tan, 2009) and Cyclin-G-associated kinase (GAK) (Dumitriu et al., 2011; Rhodes et al., 2011). These kinases, or their associated pathways, including mitogen-activated protein kinases (MAPK) such as extracellular signal-regulated protein kinases (ERK), c-Jun N-terminal kinases (JNK), p38 and protein kinase B/AKT are either risk factors associated pathological procedure or viable potential therapeutic targets (Cassarino et al., 2000; Harper and Wilkie, 2003; Silva et al., 2005; Borsello and Forloni, 2007; Karunakaran et al., 2008; Dagda et al., 2009). Other PD-associated PARK genes, such as SNCA (PARK1/4), *Parkin* (PARK2), *DJ-1* (PARK7), *GIGYF2* (PARK11) and *HTRA2* (PARK13) have been shown to function in kinase signaling related pathways (Alnemri, 2007; Schapira, 2009; Carballo-Carbajal et al., 2010; Higashi et al., 2010; Akundi et al., 2012). These further emphasize the importance of reviewing kinase encoded PARK genes (LRRK2, PINK1 and GAK) and their linked kinase pathways such as JNK,

ERK, p38 and protein kinase B/AKT signaling pathways, which could serve as potential targets for therapeutic intervention in PD.

2. Kinase genes associated with PD: Loci, physiological function and pathological role

Protein kinases catalyze protein phosphorylation, which usually results in functional changes in enzyme activity, cellular location or association with other proteins (Fig. 1). The human genome contains approximately 500 protein kinase genes, which constitute about 2% of all human genes. Up to 30% of all human proteins may be modified by kinase activity and kinases are known to function as regulators of the majority of cellular pathways and as switches in a number of cellular processes including proliferation, differentiation, survival and death (Manning et al., 2002). Dysfunction in the activity of protein kinases, such as serine/threonine-specific protein kinases, and mutations in kinase encoded genes have been implicated in the development of many human disorders including neurodegenerative diseases (Fischer et al., 2003; Das Evcimen and King, 2007; Hernandez et al., 2009; Ghigo et al., 2010; Salminen et al., 2011).

To date, with the exception of about 20 non-PARK genes without dominant association with the Parkinsonian phenotype, 18 PARK loci have been described in a series of GWAS (genome-wide association studies) (Fujioka and Wszolek, 2012; Sundal et al., 2012). Among them, at least six genes have been identified to be associated with hereditary Parkinsonian disorders. These genes are divided according to the type of inheritance into autosomal dominant (AD) or autosomal recessive (AR) and include autosomal dominant mutations in the α -synuclein (SNCA), ubiquitin-C-terminal hydrolase-L1 (UCH-L1) and the leucine-rich repeat kinase 2 (LRRK2) genes and autosomal recessively inherited mutations in *Parkin*, PTEN-induced kinase 1 (PINK1), *DJ-1* and the ATP13A2 genes (Nalls et al., 2011). The other PARK genes are either provisional or clinically associated with atypical Parkinsonism. Furthermore, GWAS for PD have linked other newly identified loci, including HLA-DRB5, BST1, GAK, ACMSD, STK39, MCCC1/LAMP3, SYT11 and CCDC62/HIP1R with risk of PD (Bonifati, 2010; Nalls et al., 2011). Among the PARK genes, there are at least three genes that encode protein kinases: PINK1 associated with autosomal recessive PD, LRRK2 associated with autosomal dominant PD with late onset and GAK with uncertain inheritance (Bonifati, 2010; Nalls et al., 2011; Fujioka and Wszolek, 2012). There are also at least two non-PARK genes that encode protein kinases: *PRKRA* and *PANK2* (Bonifati, 2010; Nalls et al., 2011; Fujioka and Wszolek, 2012) (Table 1). Moreover, there is a case-control report suggesting an association of single nucleotide polymorphisms (SNPs) in protein kinases (AKT1) with protection against PD (Xiomerisiou

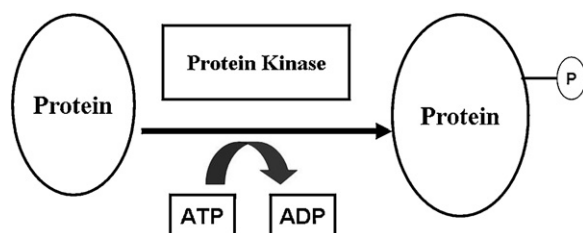


Fig. 1. General scheme of kinase function.

Table 1

Kinase genes associated with Parkinson's disease (PARK and non-PARK).

Product	Gene	Loci	Inheritance
PARK6	<i>PINK1</i>	1p36.12	AR
PARK8	<i>LRRK2</i>	12q12	AD
PARK17?	<i>GAK</i> ^a	4p16	–
DYT16	<i>PRKRA</i>	2q31.2	AR
PKAN	<i>PANK2</i>	20p13	AR

PINK1, PTEN-induced putative kinase 1; LRRK2, leucine-rich repeat kinase 2; GAK, Cyclin-G-associated kinase; DYT, dystonia; PKAN, pantothenate kinase-associated neurodegeneration; PRKRA, protein kinase, interferon-inducible double-stranded RNA-dependent activator; PANK2, pantothenate kinase 2; AD, autosomal dominant; AR, autosomal recessive? Some reports identify VPS35 (vacuolar protein sorting 35) as PARK17 (<http://www.ncbi.nlm.nih.gov/gene?term=PARK17>).

^a Provisional loci/genes identified through the recent genome-wide association studies also awaiting further confirmation.

et al., 2008) although adverse associations have been reported in more recent investigations (Ran et al., 2011). Therefore, elucidation of the normal function of LRRK2, PINK1 and other associated kinases (GAK, PRKRA, PANK2 and AKT1, et al.) and the pathological mechanisms of the mutated forms has major implications for the development of potential therapies for PD.

In this part, we will mainly discuss the loci, physiological function and pathological role of the kinases above mentioned, especially physiological substrates. Unfortunately, most kinases substrates are identified in cell culture system, and only few discovered in *in vivo* system. Therefore, more physiological substrates at the present need to be confirmed in animal and human studies in future.

2.1. PINK1 (PARK6): Loci, physiological function and pathological role

PINK1 was first identified in tumor cell expression profile experiments and localized to the PARK6 locus on chromosome 1p35–36. Because of the characteristics of transcriptional activation by PTEN, it was named PTEN-induced kinase 1 (PINK1) (Valente et al., 2001, 2004). Among the 40 or more PINK1 mutations identified in symptomatic individuals, 15 segregate with disease in pedigrees of early onset Parkinsonism (EOP). Mutations in PINK1 have been discovered in different regions including Europe, Asia and North America families with ARPD (Li et al., 2005; Zimprich, 2011). PINK1 mutations have been identified predominantly in familial cases, but interestingly, a higher incidence of heterozygous PINK1 mutations occurs in sporadic PD patients (Abou-Sleiman et al., 2006; Bonifati, 2012). Immunohistochemistry of PD brain sections demonstrated PINK1 immunoreactivity in a small proportion (5–10%) of neuronal inclusions resembling Lewy bodies (Gandhi et al., 2006), supporting its role in PD pathology.

2.1.1. Physiological function of PINK1

As a protein serine/threonine kinase, early findings indicate a function for PINK1 in intracellular signaling pathways implicated in PD pathogenesis. Three emerging properties of PINK1 are especially relevant. First, primary sequence analysis and biochemical studies have demonstrated that PINK1 is a member of the eukaryotic protein kinase family of enzymes that are involved in transduction and amplification of intracellular signals (Silvestri et al., 2005; Sim et al., 2006). Protein phosphorylation is likely to play an important role in PD pathogenesis and PINK1 may participate by directly phosphorylating or binding key cellular proteins governing neuronal survival. Second, the N-terminal region of PINK1 contains a mitochondrial targeting motif. Initial studies in mammalian cell lines suggest that this motif is sufficient to direct mitochondrial import of PINK1 (Silvestri et al., 2005;

Muqit et al., 2006). Mitochondrial localization implicates PINK1 as a potential regulator of mitochondrial processes associated with apoptosis, such as the electron transport chain and opening of mitochondrial permeability transition pores (PTPs). Studies of the function and regulation of PINK1 will therefore elucidate the role of these mitochondrial processes in the pathogenesis of familial and sporadic PD. Third, evidence from animal models and cultured mammalian cells supports a role for functional interactions between PINK1 with another PD-associated gene Parkin and the HtrA2 gene via modulation of phosphorylation in PD pathogenesis (Plun-Favreau et al., 2007; Dagda et al., 2009; Deas et al., 2011).

2.1.2. PINK1 substrates

The C-terminal regions of a number of protein kinases contain functional motifs that regulate both the catalytic activity of the kinase domain as well as binding of protein substrates and regulatory proteins (Nolen et al., 2001). The C-terminal region may direct the PINK1 kinase domain to phosphorylate specific cellular protein substrates. A recombinant protein comprising the PINK1 C-terminal region has been used as bait in experiments to identify specific binding proteins in crude mammalian brain lysates representing physiological protein substrates of PINK1 (Ghazavi et al., 2011). Several studies have reported the mitochondrial molecular chaperone tumor necrosis factor receptor-associated protein-1 (TRAP1) to be a substrate for PINK1 (Pridgeon et al., 2007). PINK1 phosphorylates TRAP1 both *in vitro* and *in vivo* and knockdown of TRAP1 decreases the expression of heme oxygenase (HO)-1 in response to oxidative stress (Chien et al., 2011). Furthermore, PINK1 mutation inhibits both upregulation of TRAP1 expression (Chien et al., 2011) and phosphorylation (Pridgeon et al., 2007) and therefore prevents protection of cells against oxidative stimuli. However, PINK1 is required for replication in other cell models, such as dopaminergic neurons, and in animal models. Furthermore, in *Drosophila*, the mitochondrial fusion-promoting factor Mitofusin (dMfn) has been shown to be a direct common substrate of the PINK1/Parkin pathway and that reduction of ubiquitin-mediated turnover of dMfn results, in part, from the mitochondrial morphological alterations and tissue degeneration phenotypes that derive from mutations in PINK1 and Parkin (Poole et al., 2010).

2.1.3. Pathological role of PINK1 in the pathogenesis of PD

Cell death of dopaminergic neurons is due to a combination of exogenous stress and genetic predisposition. The loss of PINK1 function causes mitochondrial dysfunction and increases cellular susceptibility to stress. Prior studies have indicated that PINK1 phosphorylates the downstream effector TRAP1 and the downstream interactor Omi/HtrA2 (PARK13) and plays an essential role in regulation of mitochondrial integrity in the PINK1 related pathway (Abeliovich, 2007; Pridgeon et al., 2007; Yun et al., 2008). PINK1 protects against oxidative stress-induced cell death by suppressing cytochrome c release from mitochondria via TRAP1 phosphorylation (Chien et al., 2011). In addition, PINK1 regulates the Parkin/I κ B kinase (IKK)/NF- κ B, PI3-kinase/AKT and HtrA2 signaling pathways. It has been shown that mitochondrial protease Omi/HtrA2 is phosphorylated in a PINK1-dependent manner at a residue adjacent to a position found to be mutated in PD patients upon activation of the JNK pathway. Deceased Omi/HtrA2 phosphorylation has been detected in the brains of patients with PD carrying mutations in PINK1 compared with idiopathic patients. Therefore, the interaction between the PINK1 and JNK pathway requires further investigation (Jankowski, 2007; Dagda et al., 2009; Akundi et al., 2011, 2012).

In another study, Kitada et al. (2007) revealed that PINK1 plays a critical role in dopamine (DA) release and striatal synaptic plasticity in the nigrostriatal circuit while the numbers of DA

neurons were not altered in PINK1 knockout mice. PINK1 mutations in human carriers have suggested presynaptic dopaminergic defects and loss of striatal dopaminergic terminals, which develop not only motor symptoms, but also non-motor symptoms (Weng et al., 2007; Eggers et al., 2010; Pavese et al., 2010). Additionally, PINK1 deletion leads to a functional reduction in the activity of small-conductance Ca^{2+} -activated potassium channels, and therefore causes substantia nigra compacta (SNc) dopamine neurons rather than ventral tegmental area dopamine neurons or substantia nigra pars reticulata GABAergic neurons to fire action potentials in an irregular pattern and enhances burst firing in brain slices and *in vivo* (Bishop et al., 2010). These findings suggest that PINK1 mutations are involved in the processes of selective vulnerability in SNc dopaminergic neuron in PD.

2.2. LRRK2 (dardarin, PARK8): Loci, physiological function and pathological role

LRRK2 is a large gene located on human chromosome 12 that has 51 exons and encodes a protein of 286 kDa (Cookson, 2010). Before its role in PD was reported, LRRK2-associated PD was found to present with late onset and clinical features very similar to sporadic PD, although cases with different LRRK2 mutations display variable pathology (Khan et al., 2005; Kumari and Tan, 2009). Most studies have indicated that LRRK2 mutations explain 5–13% of familial and 1–5% of sporadic PD cases. Notably, one of most common recurrent mutations (G2019S) located in the kinase domain has been found across most ethnic populations, with the highest prevalence among Ashkenazi Jews and North African Arabs (Kumari and Tan, 2009). Additionally, a common missense mutation (R1441G) in transgenic mouse model recapitulates cardinal features of PD (Li et al., 2009). LRRK2 has a modular structure that includes a series of leucine-rich repeats towards the N-terminus, and hence was named leucine-rich repeat kinase. The kinase domains of LRRK2 can be separated from most other kinases and are an offshoot of the tyrosine-like kinase family, which includes other large modular kinases involved in diverse biological functions (Berwick and Harvey, 2011). Therefore, understanding LRRK2 pathogenicity may elucidate the molecular mechanisms of sporadic PD.

2.2.1. Physiological function of LRRK2

As a large, complex, multidomain protein, LRRK2 comprises kinase and GTPase enzymatic activities and multiple protein interaction and catalytic domains. The precise cellular function of LRRK2 is still unknown. However, it has been detected in association with a variety of membrane and vesicular structures, membrane-bound organelles and microtubules with high affinity for lipids or lipid-associated proteins and may play a potential role in the biogenesis and/or regulation of vesicular and membranous intracellular structures within the mammalian brain (Biskup et al., 2006). Several studies have indicated that the LRRK2 kinase domain shares a degree of homology with MAPKKK proteins and that the LRRK proteins (LRRK1 and LRRK2) form an orphan group in the human kinome most closely related to the receptor interacting protein kinase (RIPK) family that plays critical roles in cellular stress responses (Abdalla-Carvalho et al., 2010; Gloeckner et al., 2010). However, other studies have suggested that LRRK2 requires a number of interactors, adaptor proteins and/or cofactors to carry out its normal function and that the levels of autophosphorylation observed *in vitro* may not reflect activity of the intact protein complex (Kamikawaji et al., 2009).

Overexpression of mutant LRRK2 (R1441C, Y1699C and G2019S mutants) has been implicated in enhancement of cellular toxicity in cellular models, which is consistent with the dominant inheritance pattern of this disease (Smith et al., 2006a,b; Chan

et al., 2011). Investigations of the requirement of LRRK2 kinase activity for toxicity showed that inactivation of the kinase domain *via* mutation of key residues resulted in marked amelioration of the toxicity of mutant LRRK2, even in association with a pathological mutation in the inactive kinase (Greggio et al., 2006; Greggio and Singleton, 2007). Experiments showed that mutant LRRK2 interfered with synaptic morphology and vesicular recycling was dependent upon kinase activity both *in vitro* using primary neuronal cultures and *in vivo* by delivery of the LRRK2 kinase domain (wild-type or G2019S) into rat brains mediated by adeno-associated viruses (MacLeod et al., 2006). Furthermore, silencing of LRRK2 with short hairpin RNAs led to an opposing phenotype characterized by increased neurite process length and complexity, and resulted in a redistribution of vesicles at the presynaptic site and participation in regulation of the dynamics between vesicle pools. Thus, LRRK2 protein is involved in the regulation of neurite process morphology in addition to maintenance and vesicular recycling (Greggio and Singleton, 2007; Piccoli et al., 2011).

2.2.2. LRRK2 substrates

To date, at least three LRRK2 substrates that have been reported. Firstly, Jaleel and his co-workers identified moesin, a protein that anchors the actin cytoskeleton to the plasma membrane. Related proteins (ezrin and radixin) are efficiently phosphorylated by LRRK2 at Thr558, a phosphorylation site that regulates the ability of moesin to bind actin previously identified *via* a kinase substrate tracking and elucidation (KESTREL) screen in rat brain extracts. This study suggests that moesin, ezrin and radixin are LRRK2 substrates (Jaleel et al., 2007). Other interesting observations were noted recently in two independent studies (Imai et al., 2008; Ohta et al., 2011). Imai and colleagues showed that LRRK2 phosphorylates eukaryotic initiation factor 4E (eIF4E)-binding protein (4E-BP), a negative regulator of eIF4E-mediated protein translation and a key mediator of various stress responses in both humans and *Drosophila*. Chronic inactivation of 4E-BP by LRRK2 containing pathogenic mutations deregulates protein translation, resulting, at least in part, in the pathogenic mechanisms of neurodegeneration in DA neurons (Imai et al., 2008). The latest report by Ohta et al. performed *in vitro* kinase assays with recombinant proteins to demonstrate that LRRK2 directly phosphorylates AKT1, a molecule centrally involved in signal transduction for cell survival and prevention of apoptosis by acting at the site of Ser473. This observation indicates that AKT1 is a physiological substrate of LRRK2 and that previously identified disease-associated mutations, including R1441C, G2019S and I2020T, cause neurodegeneration by reducing the interaction with, and phosphorylation of, AKT1 (Ohta et al., 2011).

2.2.3. Pathological role of LRRK2 in the pathogenesis of PD

Previous studies have demonstrated that, LRRK2, as a multidomain protein containing several protein interaction motifs as well as dual enzymatic domains with GTPase and protein kinase activities, plays an important role in LRRK2-induced neuronal toxicity and inclusion body formation. Disease-associated mutations are reported throughout the multidomain structure of the protein and tend to enhance kinase activity (Li et al., 2010; Xiong et al., 2010), which is regulated by the GDP/GTP binding state of the ROC domain. This is considered to be a small GTPase and the PD-associated mutants appear to have no effect on GTP hydrolysis activities compared with inhibition of the kinase activity (Fu et al., 2009). However, mutations in the GTPase/ROC domain have been shown to exert the opposite effects of disruption of GTP hydrolysis activity without affecting the affinity for GTP (Li et al., 2007). Therefore, these observations give rise to the intriguing scenario that modulation of GTPase activity by compound modulators may influence of LRRK2 activity. However, to date, there is little

evidence for the effective treatment of PD by GTPase inhibitors except for cancers (Walker and Olson, 2005; Fritz and Kaina, 2006). Recent investigations of the cross-regulatory activities of the kinase and ROC domains revealed that autophosphorylation of LRRK2 mainly targets the ROC GTPase domain and that LRRK2 clusters around the GTP binding pocket of the ROC domain (Gloeckner et al., 2010). This study suggested that the phosphorylation of LRRK2 regulates its biological functions via a mechanism that remains to be elucidated. In addition, Lin and colleagues found that overexpression of LRRK2 induced cell death via promoting the abnormal aggregation and somatic accumulation of α -synuclein and the inhibition of LRRK2 expression could ameliorate α -synuclein-induced neurodegeneration (Lin et al., 2009). The results reveal an interesting interplay between α -synuclein and LRRK2, two gene products associated with dominantly inherited PD (Tong and Shen, 2009).

As an most important determinant of PD, the G2019S mutation has been shown to be responsible for upregulation of LRRK2 kinase activity, which may ultimately lead to neuronal loss, and is the most frequent genetic cause of PD (Kumari and Tan, 2009; Benamer and de Silva, 2010). Previously, a chemical library screen identified inhibitors of LRRK2 kinase activity. Two of these inhibitors, GW5074 and sorafenib, protect against G2019S LRRK2-induced neurodegeneration *in vivo* in *Caenorhabditis elegans* and in *Drosophila* (Liu et al., 2011). These findings indicate that increased LRRK2 kinase activity is neurotoxic and that inhibition of LRRK2 activity may have a disease-modifying effect (Liu et al., 2011). Therefore, developing selective inhibitors for the kinase activity of LRRK2 may have potential therapeutic benefit in PD.

More recently, a relatively large-scale chemical library screen revealed a panel of inhibitors of LRRK2 kinase activity in addition to those previously mentioned (Lee et al., 2010). Deng et al. (2011) identified LRRK2-IN-1, a potent and selective inhibitor of LRRK2-induced dephosphorylation of Ser910 and Ser935 and accumulation of LRRK2 within aggregate structures. Furthermore, Yun et al. (2011) identified another novel potential LRRK2 inhibitor among these compounds. Therefore, it can be speculated that LRRK2 inhibitors may represent promising compounds for the development of PD therapeutics.

2.3. Potential others (cyclin-G-associated kinase (GAK), PARK17? and Tau kinases): Loci, physiological function and pathological role

Other novel PARK and non-PARK genes have been identified recently in a series of GWAS investigations. However, some of these are controversial and require further confirmation. For example, different studies provide support for VPS35 or GAK as PARK17 and HLA or EIF4G1 as PARK18. Furthermore, besides the PARK genes, two non-PARK genes, PRKRA and PANK2, have been shown to encode kinase products. However, the clinical phenotype of non-PARK genes is usually dominated by other symptoms such as ataxia, chorea, dystonia and/or dementia (Fujioka and Wszolek, 2012; Sundal et al., 2012). In addition, two tau kinases, including glycogen synthase kinase-3 β (GSK3 β) and cyclin-dependent kinase 5 (CDK5), are thought of playing key roles in the pathogenesis of PD via involving the abnormal phosphorylation of tau (Camins et al., 2006; Das et al., 2012; Kwok et al., 2005). Therefore, in this review, the newly discovered PARK gene-GAK and tau kinase (GSK3 β and CDK5) are discussed as a representative of other kinases associated with PD.

2.3.1. GAK: Loci, physiological function and pathological role

2.3.1.1. Physiological function of GAK. As a ubiquitously expressed 160 kDa serine/threonine kinase, GAK has been shown to participate in membrane trafficking (Kanaoka et al., 1997; Kimura et al., 1997; Greener et al., 2000; Korolchuk and Banting, 2002), and

to localize in both the cytoplasm and nucleus (Sato et al., 2009). Early findings revealed that GAK was involved in both the ultimate disassembly of clathrin-coated vesicles via its auxilin homology domains and in their formation through its kinase domain (Conner et al., 2003; Zhang et al., 2004). Furthermore, Shimizu et al. (2009) observed cell-cycle arrest at metaphase, and multi-aster formation, which result from abnormal fragmentation of pericentriolar material in GAK-siRNA-treated cells. Their results revealed two novel functions of GAK: maintenance of proper centrosome maturation and of mitotic chromosome congression (Shimizu et al., 2009). Furthermore, Tanenbaum et al. demonstrated that GAK acts through clathrin to control microtubule outgrowth around chromosomes (Tanenbaum et al., 2010). In addition to its characteristic role in formation of complexes with nuclear proteins such as cyclin G1, p53, clathrin heavy chain (CHC) and protein phosphatase 2A (PP2A) B'γ1, there is evidence of a novel role for GAK in regulation of dephosphorylation events under the control of the PP 2A B'γ subunit (Naito et al., 2012).

2.3.1.2. GAK substrates. To date, few studies have reported substrates for GAK, with only one report identifying the medium subunit of the AP2 adaptor complex (mu2) as a substrate for GAK/auxilin2 in studies of highly purified clathrin-coated vesicles (CCVs) isolated from porcine brain (Korolchuk and Banting, 2002). It is well-known that, as part of the clathrin adaptor complex, the AP2 complex is involved in endocytosis at the plasma membrane and comprises four different subunits: α (large), β 2 (large), μ 2 (medium) and σ 2 (small) (Matsui and Kirchhausen, 1990). In particular, mu2 recruits cargo proteins with specific tyrosine-based Yxx ϕ motifs and partly mediates the association of the AP2 complex with membranes (Owen and Evans, 1998; Honing et al., 2005). Recently, Gu and colleagues investigated the *apm-2* gene (also called *dpy-23*), which encodes the only mu2 in the nematode *Caenorhabditis elegans*, and found that loss of APM-2 function in neurons has no effects on locomotion. This suggests that Mu2 adaptin is involved in, but not essential for, synaptic vesicle recycling in the fly (Gu et al., 2008). Further investigations involving substrate tracking systems are expected to reveal novel physiologic substrates in the future.

2.3.1.3. Pathological role of GAK in the pathogenesis of PD. Previous studies of the vital role of GAK in cell proliferation and receptor trafficking have revealed that GAK represents a prognostic marker in advanced diseases, such as hormone refractory prostate cancer and osteosarcoma (Ray et al., 2006; Susa et al., 2010). In contrast, replication studies of GWAS associations for GAK and MAPT in PD in 2011 implicated GAK as a novel candidate therapeutic target for PD (Rhodes et al., 2011). In population-based studies, Rhodes et al. (2011) provided evidence for replication of rs1564282 in GAK ($OR = 1.61$, $P = 0.0151$) accompanied by replication of rs1724425 near MAPT ($OR = 0.74$, $P = 0.0163$) by using identical analytical techniques and genetic models in independent samples. Almost simultaneously, another group in the USA reported significant association of rs1564282 in GAK with increased PD-risk based on meta-analysis of three publicly available PD case-control GWAS and genotyping from a new, independent Italian cohort. Furthermore, studies conducted in cellular models of PD revealed that GAK inhibition enhanced toxicity accompanied by overexpression or A53T mutation of α -synuclein (PARK1/4). These observations provide evidence that GAK modification of α -synuclein expression levels and toxicity is involved in PD pathogenesis (Dumitriu et al., 2011). Although, details of the role of GAK in the pathogenesis of PD are undetermined, it is expected that current information will provide the basis of further studies of GAK and other pathways (Dumitriu et al., 2011; Rhodes et al., 2011), which may provide therapeutic targets for the treatment of PD.

2.3.2. Tau kinases (*GSK3 β* and *CDK5*): Loci, physiological function and pathological role

2.3.2.1. Physiological function of tau kinases.

2.3.2.1.1. *GSK3 β* . *GSK-3* is mapped to 3q13.3 (Shaw et al., 1998), and it encodes a serine/threonine kinase in charge of phosphorylating the rate-limiting enzyme-glycogen synthase, a central molecule mediating several signaling pathways involved in the regulation of cell polarization, mitosis and apoptosis, protein synthesis, and glycogen metabolism, (Kockeritz et al., 2006). *GSK3* has two similar isoforms in mammals, *GSK3 α* and *GSK3 β* , which are extensively expressed and differ in their N- and C-terminal regions (Soutar et al., 2010; Woodgett, 1990). Exonal deletions of *GSK3 α* and *GSK3 β* genes generate distinct phenotypes, differing in the developmental profiles and its molecular functions, as the two isoforms are not identical and have a different spectrum of substrates (Soutar et al., 2010). *GSK3 β* is a multi-task kinase that plays important roles in many signaling pathways, such as Wnt and Hedgehog signaling pathways, and JNK pathways (Kaidanovich-Beilin and Woodgett, 2011; Liu et al., 2004). *GSK3 β* serves as a physiological regulator in cell division, stem cell renewal and differentiation, apoptosis, circadian rhythm and insulin secretion (Lei et al., 2011).

2.3.2.1.2. *CDK5*. *CDK5* gene is mapped to 7q36 (Demetrick et al., 1994), encoding a proline-directed serine/threonine kinases ubiquitously expressed in nervous system and non-nervous system. There are now substantial data indicating that *CDK5* not only plays an important role in regulation of neuronal migration, cellular skeleton, axonal guidance, and synaptic plasticity, but also in modulations of gene transcription, vesicular transport, apoptosis, cell adhesion and migration in non-neuronal cells and tissues such as pancreatic cells, myocytes, neutrophils and others (Rosales and Lee, 2006). Unlike the other members of the *CDK* family, *CDK5* is not directly contributed to cell cycle regulation, being normally linked to neuronal activity, including neuronal migration, cortical development and synaptic plasticity (Lopes and Agostinho, 2011). This kinase is present mainly in post-mitotic neurons and its activity is regulated by the interaction with neuron-specific activator molecules (p35, p39 and p67) rather than cyclins themselves (Amin et al., 2002).

2.3.2.2. Tau kinases substrates.

2.3.2.2.1. *GSK3 β* . There are increasing bodies of research demonstrate that *GSK3 β* phosphorylates a number of microtubule-associated proteins, including Tau, MAP1B, MAP2, APC, CRMP2, to modulate microtubule growth and stability and inhibit their binding ability with microtubules (Cohen and Frame, 2001; Cole et al., 2006; Goold and Gordon-Weeks, 2001; Jope and Johnson, 2004; Kosik and Shimura, 2005; Rubinfeld et al., 1996; Sanchez et al., 2000). In addition, Cheng and co-workers identified the spindle-associated protein Astrin as an interaction partner and candidate substrate for *GSK3 β* and found that *GSK3 β* and Astrin are linked at the spindle apparatus during mitosis *in vitro* (Cheng et al., 2008). Furthermore, one recent study also identified the brain-enriched cytoskeleton-associated protein β -adducin as a novel substrate of *GSK3* according to bioinformatics and biochemical validation with confirmation of *in vitro* experiments (Farghaian et al., 2011). The study found that phosphorylation of β -adducin by *GSK3* promotes efficient neurite outgrowth in neurons (Farghaian et al., 2011).

2.3.2.2.2. *CDK5*. Firstly, *CDK5* contributes to mossy fiber sprouting (MFS) via phosphorylating its substrate tau protein, which was known to facilitate microtubule stabilization and axonal elongation (Tian et al., 2010). Secondly, *CDK5* can either increase or decrease neurotransmitter release by phosphorylating various proteins implicated in exocytosis of synaptic vesicles and insulin action, including synapsin I (Rosales et al., 2000), Munc18

(Shimomura et al., 2011), P/Q type voltage-dependent Ca^{2+} -channels (Tomizawa et al., 2002) and β 2-syntrophin in neurons (Schubert et al., 2010). Among them, the stimulatory and inhibitory roles of *CDK5* on insulin secretion have been attributed to the phosphorylation of Munc18 (Lilja et al., 2004), and a serine in the α 1 subunit of the L-type voltage-gated Ca^{2+} -channels (Wei et al., 2005), respectively. In addition, recently Wong AS and colleagues identified endophilin B1 (EndoB1) as a *CDK5* substrate, and found that *CDK5*-mediated phosphorylation of EndoB1 is required for autophagy induction in starved neurons (Wong et al., 2011). Furthermore, *CDK5*-mediated phosphorylation of EndoB1 is essential for autophagy induction and neuronal death in PD models (Wong et al., 2011).

2.3.2.3. Pathological role of tau kinases in the pathogenesis of PD. Previous investigations indicated that there is a potential role of tau emerging in PD, then pathological role of *GSK3 β* and *CDK5* in the pathogenesis of PD has also been discussed, respectively.

2.3.2.3.1. *GSK3 β* . PD associated neurotoxins like MPTP and paraquat involved in the pathways that activate *GSK3 β* and influence levels of *GSK3 β* in the basal ganglia (Lucas et al., 2001; Songin et al., 2011), and this finding has been confirmed in postmortem brain from PD patients (Duka et al., 2009). Furthermore, polymorphisms in *GSK3 β* , which affect its transcription and splicing, exert synergistic effect with another oxidative stress-related genes (heme oxygenase-1) (Infante et al., 2010), *CDK5* is also modified disease risk in PD from research units of Australia (Kwok et al., 2005), Spain (Infante et al., 2010), Greek (Kalinderi et al., 2011) and eastern India (Das et al., 2012), respectively. Additionally, there are studies to reveal an interaction between *GSK3 β* and α -synuclein in the pathogenesis of PD. As a potential substrate of *GSK3 β* , α -synuclein may also modulate the activation of *GSK3 β* (Duka et al., 2009), in one *in vitro* study, Kawakami and co-worker found that: α -synuclein directly stimulates the phosphorylation of tau by *GSK3 β* , and forms a heterotrimeric complex with tau and *GSK3 β* , they concluded that α -synuclein functions as a connecting mediator for tau and *GSK3 β* , resulting in *GSK3 β* -mediated tau phosphorylation (Kawakami et al., 2011). Therefore, α -synuclein lacking will result in abolishment of *GSK3 β* phosphorylation at Tyr216, and the inhibition of *GSK3 β* protects against MPTP toxicity *in vitro* and *in vivo* and decreases α -synuclein protein expression (Duka et al., 2009; Kozikowski et al., 2006; Wang et al., 2007). In addition, the neuroprotective possibilities of *GSK3* inhibition on tau and α -synuclein pathology have thus been suggested that small molecules *GSK3 β* inhibitor rather than lithium may offer an important strategy in the treatment of PD in future.

2.3.2.3.2. *CDK5*. Calpain mediated *CDK5* activation is critical for mitochondrial cytochrome-induced dopaminergic death in MPP⁺ induced cellular model and MPTP-induced mouse model of PD (Smith et al., 2003, 2006a,b). More importantly in parallel, increased *CDK5* level has been identified in human postmortem PD brains (Brion and Couck, 1995; Nakamura et al., 1997). During the loss of dopaminergic neurons, the calpain-p35-p25/*CDK5*-mediated activation plays a very important role (Smith et al., 2006a,b). *CDK5* is enzymatically inactive unless bound to its neuron-specific activator molecules, p35, then, inappropriate activation of the *CDK5*/p35 signal leads to neuronal death through pathogenic activation of calpains, thereby converted P35 to a pathogenic P25. This activator switching will cause a shift in the phosphorylation pattern of *CDK5*, with an alteration both in targets and activity, ultimately leading to neuronal loss (Amin et al., 2002; Ko et al., 2001; Smith et al., 2006a,b). Recently rs735555 of *CDK5* regulatory subunit (*CDK5R1*) that codes for p35, the rate-limiting factor for *CDK5* expression, has been shown to have almost twice the risk of developing PD than that without this genotypic

combination with GSK3 β (rs6438552), thereby increasing the risk of PD in an India cohort (Das et al., 2012). In addition, CDK5 has associated with the development of levodopa induced dyskinesia (LID). Guan et al. (2010) have examined CDK5 inhibitor, roscovitine in LID rats and found that the induction of LID upregulated the expression of CDK5 and phosphorylation of DARPP-32 at Thr34 and roscovitine not only improved the behaviors, but downregulated the expression of CDK5 in LID rats. These findings suggest that modulation of CDK5 may have potential therapy of human PD in future.

3. Kinase signaling pathways associated with PD: Characteristics, physiological function and pathological role

LRRK2, PINK1 and other important PARK gene products such as SNCA (PARK1/4), Parkin (PARK2) and DJ-1 (PARK7) have been shown to function in kinase signaling related pathways, including MAPK (such as ERK, JNK and p38) and protein kinase B/AKT (Alnemri, 2007; Schapira, 2009; Carballo-Carbajal et al., 2010; Higashi et al., 2010; Akundi et al., 2012). These signaling pathways modulate mitochondrial respiration, transport, fission-fusion, calcium buffering, reactive oxygen species (ROS) production, autophagy and apoptotic cell death. In the next part of this review, we summarize accumulating experimental evidence, gathered over the last decade, of a central role for kinase signaling and interactions involving LRRK2, PINK1, DJ-1 and other proteins in PD. Converging mechanisms from different model systems support the concept of common kinase signaling pathways in Parkinsonism that may be amenable to potential therapeutic targeting in future.

3.1. Mitogen-activated protein kinase signaling pathway

The MAP kinase cascade consists of three enzymes, MAP kinase, MAP kinase kinase (MKK, MEK or MAP2K) and MAP kinase kinase kinase (MKKK, MEKK or MAP3K), which are activated in series (Pearson et al., 2001). Accumulating evidence has shown that the MAP kinase signaling cascade is involved a variety of physiological responses to extracellular stimuli, such as mitogens, osmotic stress, heat shock and proinflammatory cytokines, and regulates various cellular activities, such as gene expression, mitosis, differentiation, proliferation and cell survival/apoptosis (Pearson et al., 2001). Among the subfamily of MAP kinases, Jun N-terminal kinases/stress-activated protein kinases (JNK/SAPKs), ERK and p38 are believed to play a role in the pathogenesis of PD (Greggio and Singleton, 2007; Dagda et al., 2009).

3.1.1. c-Jun N-terminal kinase (JNK) signaling pathway and PD

3.1.1.1. JNK signaling pathway and JNK inhibitors in PD. There is evidence that the JNK signaling cascade regulates the death program of neuronal cells and may represent a suitable therapeutic target for rational drug design in PD. JNK promotes neuronal death by regulating the activation of downstream pathways, such as the nuclear and non-nuclear pathways. Activated JNK regulates various nuclear substrates, such as c-Jun, as well as cytosol substrates. In fact, previous studies suggest that c-Jun plays an important role in neuronal death under *in vitro* and *in vivo* conditions (Pei et al., 2006). Activated JNK phosphorylates the transcription factor c-Jun and the subsequent increase in activating protein-1 (AP-1) activity modulates the transcription of a number of genes such as Fas ligand (Fas L) (Faris et al., 1998). Apart from the nuclear pathway, JNK also promotes cell death by regulating the activation of non-nuclear substrates, such as Bcl-2 family members. Recent studies have indicated that after activation by 6-hydroxydopamine (6-OHDA), JNK3 enhances the phosphorylation of 14-3-3 protein and

promotes Bax translocation to the mitochondria, causing cytochrome c release and increasing caspase-3 activation. Furthermore, the cascades associated with 6-OHDA were shown to induce the neurodegeneration in PD (Pan et al., 2007, 2009) (Fig. 2).

In recent years, JNKs and their upstream activators have become especially attractive as drug targets. As a result, a series of prominent agents targeting JNK signaling, including upstream kinase inhibitors (for example, CEP-1347), small chemical inhibitors of JNK (SP600125 and AS601245) and peptide inhibitors of the interaction between JNK and its substrates (D-JNKI and I-JIP) have been investigated for potential roles in PD (Kuan and Burke, 2005). The small-molecule inhibitors of the MLK family of kinases, CEP-11004 and CEP-1347, prevent the activation of the JNK pathway and, consequently, decrease neuronal cell loss (Peng et al., 2004; Wang and Johnson, 2008; Wang et al., 2009). In preclinical models of PD, CEP-1347 has been shown to inhibit activation of the JNK pathway and cell death in many cell culture and animal models. More recent studies have shown that K252a, an analog of CEP-1347, inhibits the MLK3 pathway and the death effector, caspase-3. The small neuropeptide inhibitor, Tat-JBD (JNK-binding domain) significantly prevented 6-OHDA/1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP)-induced nigral dopaminergic injury (Pan et al., 2007, 2010). SP600125 is an ATP-competitive inhibitor that directly targets and potently inhibits JNK and has been shown to protect dopaminergic neurons in 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) models (Wang et al., 2009). However, the clinical failure CEP-1347 for PD led to reassessment of the relevance of both apoptosis in the etiopathogenesis of neurodegenerative diseases and of the available animal models (Waldmeier et al., 2006; Wang and Johnson, 2008). This controversy indicates the necessity for further clinical trials of

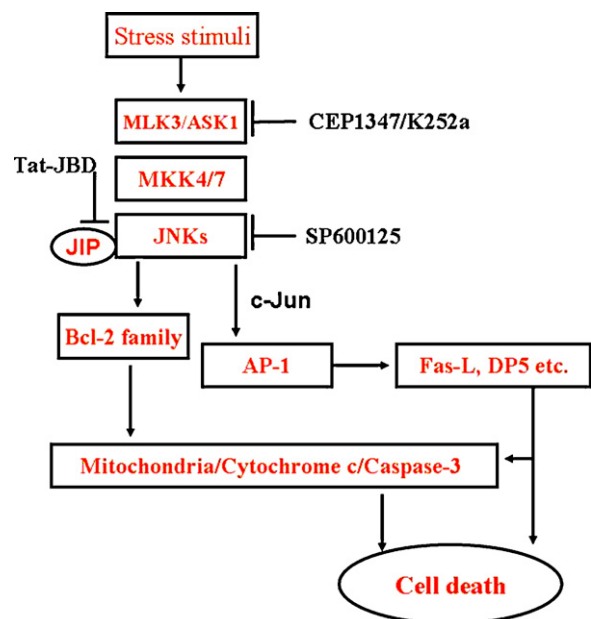


Fig. 2. The JNK signaling pathway in neuronal cell death and the sites of actions of pharmacological inhibitors. Activation of the JNK pathway represents an important step in the pathways leading to neuronal cell death in response to a variety of extracellular stimuli. Activation of JNK initiates the phosphorylation and induction of the transcription factor c-Jun, thus triggering the expression of a number of apoptosis-regulatory proteins. Activated JNKs also promote apoptosis by inactivating suppressors of the mitochondrial-dependent death pathway, releasing inhibition of apoptosis and, thus releasing pro-apoptotic molecules including cytochrome c from mitochondria. This leads to caspase activation and cell death. CEP-1347 and K252a inhibit the signal cascade at the level of MLK3. SP600125 directly inhibits JNK by an ATP-competitive mechanism. The JNK-binding domain (JBD) of JIP-1 competes with the protein target for binding to JNK. Antisense oligodeoxynucleotides to JNK or c-Jun can also be used to inhibit the expression of these proteins and to block the amplification actions.

chemical neuroprotective agents identified in preclinical experiments regardless of previous results.

3.1.1.2. Interactions between the JNK pathway and LRRK2. Interestingly, the kinase domain of LRRK2 most closely resembles kinases of the MLK (mixed lineage kinase) family. LRRK2 shares highest sequence homology with MLKs (mixed lineage kinases), which act upstream of canonical MAPKK and are involved in cellular stress responses. Recently, several researchers have demonstrated that LRRK2 variants phosphorylate MAPKK, including MKK3, -4, -6 and -7, which act upstream of the MAPK p38 and JNK (Gloeckner et al., 2009; Hsu et al., 2010a,b). Therefore, the PD-associated G2019S and I2020T mutations show an increased phosphotransferase activity towards MKKs and provide direct evidence of the correlation of MKKs with mutant LRRK2 linked to neurotoxicity, cellular stress and apoptosis. In addition, LRRK2 binds the JNK-interacting proteins (JIPs)1–4, which are a group of scaffold proteins that play an important role in the regulation of MAP kinase signaling cascades (Hsu et al., 2010b). Thus, these results demonstrate a putative interaction between LRRK2 and the MAP kinase pathway in regulating the stress kinase cascade.

3.1.1.3. Interactions between JNK pathway and PINK1. Recently, details of the JNK pathway have emerged that could link malfunction of mutated PINK1 and other proteins to cell stress-induced apoptosis. Park and co-workers found that the JNK target gene *puc* is transcriptionally activated in the muscle of PINK1-deficient *Drosophila* and that the mitochondrial impairment-induced apoptosis resulting from PINK1 ablation is mediated by ectopic activation of the JNK pathway (Park et al., 2006). Furthermore, Akundi et al. (2011, 2012) observed that PINK1 deletion in mice resulted in the increased mitochondrial Ca^{2+} sensitivity and JNK activity that precede reduced DA levels and abnormal DA homeostasis. Increasing evidence show that PINK1 regulates the localization of Parkin to the mitochondria in its kinase activity-dependent manner, and at the same time, Parkin suppresses JNK signaling by reducing *bsk* transcription in *Drosophila* (Kim et al., 2008; Hwang et al., 2010). Therefore, further investigations should focus on deciphering how PINK1 influences the JNK pathway via Parkin regulation.

3.1.2. Extracellular signal-regulated kinase (ERK) signaling pathway and PD

3.1.2.1. ERK signal pathway and ERK inhibitors in LID. The ERK signaling pathway forms part of striatal cellular signaling network and is associated with the effects of drugs abuse in the striatum (Brambilla, 2003). Furthermore, in the dopamine-depleted striatum, ERK signaling plays an important role in the development of L-DOPA-induced dyskinesia (LID) both in mouse and non-human primate models of PD (Pavon et al., 2006; Santini et al., 2007; Santini et al., 2010). A series of cascades have been proposed to be involved in the pathogenetic mechanisms of LID, including upstream (sensitized cAMP/cAMP-dependent protein kinase/DARPP-32 signaling), axis (phosphorylation/activation of ERK) and activation of two downstream targets (mitogen- and stress-activated kinase-1 (MSK-1) and phosphorylation of histone H3). These cascades, in turn, modulate transcriptional regulation (Fig. 3). Santini showed that L-DOPA activates ERK signaling specifically and phosphorylates histone H3 in the striatonigral medium spiny neurons (MSNs) in dyskinetic mice. These effects were blocked by dopamine D(1) receptors and were associated with ERK-dependent changes in striatal plasticity leading to dyskinesia (Santini et al., 2009).

Recently, inhibition of ERK1/2 by SL327 (alpha-[amino[(4-aminophenyl)thio]methylene]-2-(trifluoromethyl)benzeneacetonitrile), an inhibitor of the mitogen-activated kinase/ERK kinase,

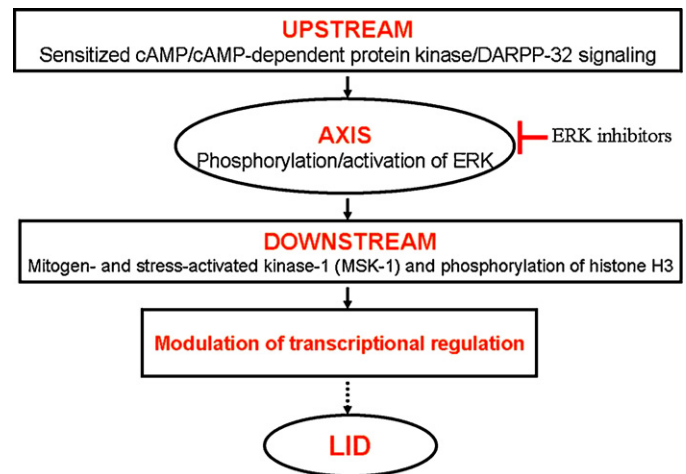


Fig. 3. The pathogenetic mechanisms of L-DOPA-induced dyskinesia (LID). It has been proposed that a series of special cascades and improving or modulating ERK signaling may be potentially therapeutic effects in LID.

attenuated LID and completely inhibited all markers of angiogenesis in rat and mouse models of LID (Santini et al., 2007; Lindgren et al., 2009). Therefore, it is believed that improving or modulating ERK signaling in response to dopamine in PD patients may have potentially therapeutic effects for the cardinal signs and motor complications. Unfortunately, there are few reports describing ERK expression in the brain tissue of PD patients and preclinical trials for ERK inhibitors require further investigation in non-human LID models.

3.1.2.2. Interactions between ERK pathway and LRRK2. Notably, ERK plays a key role in cross-talk between two dominant PD genes, LRRK2 and SNCA. Overexpression of wild-type LRRK2 in human embryonic kidney HEK293 cells selectively activated the ERK signaling pathway and induction of the ERK module by LRRK2 was related to a small but significant induction of SNCA, which was inhibited by the selective MAPK/ERK kinase inhibitor 1,4-diamino-2,3-dicyano-1,4-bis[2-aminophenylthio]butadiene (U0126) (Carballo-Carbajal et al., 2010). Furthermore, U0126 attenuated LRRK2-induced neuritic autophagy and reduced neurite shortening, indicating that MAPK/ERK-related signaling mediated LRRK2 pathogenic mutation-induced autophagy in neurite remodeling (Plowey et al., 2008).

3.1.2.3. Interactions between the ERK pathway and DJ-1. The integral mitochondrial protein, DJ-1 is associated with oxidative stress and mutations in DJ-1 are involved in an autosomal recessive, early onset familial form of PD (Tomiya et al., 2009). Compelling evidence revealed that DJ-1 influences the ERK1/2 signaling pathway and results in alterations in the susceptibility of cells to oxidative stress (Gu et al., 2009; Lev et al., 2009). Overexpression of DJ-1 increased dopamine cell resistance to toxicity and reduced intracellular ROS production. Upregulation of DJ-1 was mediated by the MAP kinase pathway through activation of ERK 1, 2 *in vitro* and *in vivo* (Gu et al., 2009). Our data showed that DJ-1 exerts antioxidative functions that regulate SOD1 expression through the ERK1/2-ELK1 pathway in its protective response to oxidative insult (Wang et al., 2011). A close correlation was also revealed between the levels of DJ-1 and SOD1 in salivary samples of patients with PD (Wang et al., 2011). Furthermore, DJ-1 upregulated tyrosine hydroxylase expression by activation of its transcription factor Nurr1 via the ERK1/2 pathway. Therefore, it can be speculated that the protective effects of DJ-1 associated with PD are ERK-dependent.

3.1.3. p38 signaling pathway and PD

3.1.3.1. p38 signal pathway and p38 inhibitors for PD. Mammalian p38 MAPKs are activated by various cellular stresses, as well as in response to inflammatory cytokines. Activation of this cascade promotes cell apoptosis, a process known to be involved in PD, and plays an important role in the mechanism of neuroinflammation (Wilms et al., 2003; Karunakaran and Ravindranath, 2009; Yasuda et al., 2011). Increasing evidence from experiments *in vivo* and *in vitro* suggests that selective phosphorylation of p38 MAP kinase within the dopaminergic neurons induced by neurotoxins (such as 6-OHDA, MPTP, Rotenone et al.) promotes cell apoptosis by activation of downstream pathways, including the proinflammatory transcription factor nuclear factor kappaB (NF-kappaB) and p53. These two main downstream molecules induce upregulation of inflammatory cytokines (including tumor necrosis factor alpha, interleukin-6 and nitric oxide) and activation of mitochondrial apoptotic pathways (including transcription of Bax and Puma and activation of Caspase-8 and Caspase-9), respectively (Choi et al., 2004; Newhouse et al., 2004; Karunakaran et al., 2008; Karunakaran and Ravindranath, 2009). Importantly, increased staining of phosphorylated p38 was observed in the surviving neurons of Substantia Nigra Pars Compacta (SNpc) in human brain sections from PD carriers and suggested a role for p38 in the degeneration of dopaminergic neurons of SNpc (Karunakaran et al., 2008). In addition, endogenous factors initiating microglial activation also may depend on activation of p38. Wilms et al. (2003) showed that human neuromelanin promotes microglial activation *via* p38 MAPK in rat microglia *in vitro*.

Due to its important role in apoptotic pathways in neurons, p38 MAPK has become a target for novel drug development for neurodegenerative diseases, such as PD. Early animal transplant experiments have shown that inhibitors of p38 MAP kinase increase the survival of transplanted dopaminergic neurons (Zawada et al., 2001). Other studies have also shown that p38 MAPK inhibitors (SB239063, PD169316) protect cells against the neurotoxicity induced by different insults involved in PD (Choi et al., 2004; Wang et al., 2005; Karunakaran et al., 2008; Karunakaran and Ravindranath, 2009). However, although extensive preclinical studies and clinical trials in inflammatory diseases have been reported, few have been conducted in PD and progress in this area is slow (Yong et al., 2009, 2011). Thus, further research into the applications of p38 MAPK inhibitors as therapeutic agents in PD is warranted.

3.1.3.2. Interaction between the p38 pathway and LRRK2. As previously discussed (Interactions between the JNK pathway and LRRK2), LRRK2 variants phosphorylate MAPKK, including MKK3, -4, -6 and -7, which act upstream of MAPK p38 and JNK (Gloeckner et al., 2009; Hsu et al., 2010a,b). Among these, the MLK family of kinases activate the p38 pathway by phosphorylating MKK3 and MKK6. These MKKs are then recruited into a multi-protein complex by scaffold proteins JIPs (Gallo and Johnson, 2002). Recently, Hsu et al. elegantly demonstrated that the disease-linked LRRK2 mutations, G2019S, R1441C and I2020T, enhance binding of LRRK2 to MKK6 and LRRK2-mediated protection was abolished by p38 knockdown through *in vivo* studies in *C. elegans* (Hsu et al., 2010a,b). Thus, these data demonstrate that the interactions between the p38 pathway and LRRK2 *via* MKKs play an important role in modulating the cellular stress response.

3.1.3.3. Interactions between the p38 pathway and DJ-1. In addition to mediating direct protection in neurons, DJ-1 regulates neuroinflammatory signaling in glia. Both DJ-1 knockout *C. elegans* *in vivo* and DJ-1-knockout mouse astrocytes *in vitro* showed

stronger phosphorylation of p38 MAPK with generation of more inflammatory cytokines compared with intact controls (Waak et al., 2009; Cornejo Castro et al., 2010). Furthermore, DJ-1 regulates the p38 MAPK pathway through physical interaction with apoptosis signal-regulating kinase 1 (ASK1). Mo and colleagues found that overexpression of DJ-1 inhibited H₂O₂-induced activation of ASK1, as well as the activation of downstream kinases in the p38 MAPK signaling cascade. However, DJ-1 has shown direct effects on ASK1 rather than either MKK3 or p38 (Mo et al., 2010). Thus, these results suggest that the interactions between the p38 pathway and DJ-1 are mediated *via* ASK1 signaling cascades and are involved in cellular stress.

3.2. AKT (protein kinase B) signaling pathway

As a key mediator of signaling downstream of PI3 kinase, AKT (also known as protein kinase B, PKB), is a serine/threonine protein kinase involved in multiple cellular processes, such as regulation of cell survival, proliferation and insulin-dependent metabolic cell responses. It is composed of at least three subunits (AKT1/PKB α , AKT2/PKB β and AKT3/PKB γ) in humans (Song et al., 2005; Fayard et al., 2010). Studies conducted over the last two decades have firmly established that AKT executes these diverse tasks through phosphorylation of numerous cellular substrates and plays an important role in physiological and pathological settings. Aberrant AKT signaling in human diseases is emphasized in diabetes, cancer and neurological diseases (Vasudevan and Garraway, 2010). Significant attention has been drawn to the potential role of disrupted signaling pathways (PI3K/AKT, mTOR, eIF4/p70S6K and Hif-1 α) that comprise a network involved in the control of processes including energy metabolism and cell survival in response to growth factors, oxidative stress and nutrient deprivation. Among these, PI3K/AKT and mTOR signaling are considered as central hubs of this network, which is of relevance to longevity and induction of mitochondrial biogenesis (Elstner et al., 2011). Therefore, targeting the PI3K-AKT pathway, which is a major neuronal prosurvival pathway in response to oxidative stress, may represent a novel target for treatment of human neurodegenerative diseases.

3.2.1. The PI3K/AKT signaling pathway and PD

Compelling evidence suggests that defective PI3-kinase-AKT (PKB) signaling is associated with loss of dopaminergic neurons in PD and that activation of AKT *via* reconstruction by AKT/Rheb-induced axon regeneration may provide neuroprotection in PD (Timmons et al., 2009; Kim et al., 2011). Double immunofluorescence microscopy studies of the SNpc have demonstrated a significant decrease in expression of AKT and phosphor (Ser473)-AKT in tyrosine hydroxylase (TH) immunoreactive dopaminergic neurons in the brain of PD patients compared with control human brain tissue (Timmons et al., 2009). Furthermore, interestingly, histological analysis reported by Kim et al. (2011) demonstrated regrowth of axons from dopaminergic neurons to their target, the striatum, after adeno-associated virus (AAV)-mediated transduction of surviving endogenous dopaminergic neurons of the substantia nigra (SN) with constitutively active forms of the AKT and the GTPase, Ras homolog enriched in brain (Rheb). Some neuroprotective agents, such as 2,3,5,4'-tetrahydroxystilbene-2-O- β -D-glucoside (TSG, an active component of the rhizome extract from *Polygonum multiflorum*), caffeine and insulin-like growth factor-I (IGF-1), have been shown to afford significant anti-apoptotic effects in cellular models *via* activation of PI3/AKT pathways. These effects were blocked by LY294002 (an inhibitor of PI3-kinase) (Nakaso et al., 2008; Cheng et al., 2011; Qin et al., 2011). The activation of this signaling pathway may represent a novel and promising therapeutic target for PD.

3.2.2. Interaction between the PI3K/AKT signaling pathway and LRRK2, PINK1

As previously discussed (LRRK2 substrates), AKT1 is a convincing candidate for the physiological substrate of LRRK2. Reduction of the interaction between PI3K/AKT signaling pathway and LRRK2 caused by the disease-associated mutations, R1441C, G2019S and I2020T, are involved in the pathogenetic mechanism of PD (Ohta et al., 2011). Furthermore, the IGF-1/AKT signaling plays an important role in mediating neuronal survival by inhibiting dopamine neuron loss in several animal models of PD and both IGF-1 and insulin are neuroprotective in various settings. Akundi et al. (2012) generated PINK1-deficient cells and investigated the roles of PINK1 in IGF-1 and insulin-dependent phosphorylation of AKT and the regulation of downstream AKT target proteins. PINK1 was shown to be required for optimal IGF-1 and insulin-dependent AKT signaling. This raises the possibility that defective IGF-1/AKT signaling is involved in PINK1-related degeneration in PD by increasing the vulnerability of dopaminergic neurons to stress-induced cell death (Akundi et al., 2012). A recent novel study by Sanchez-Mora et al. (2012) provides direct evidence for interactions between PI3K/AKT and PINK1. It was shown that PINK1 overexpression prevents mitochondrial dysfunction, reinforces anti-apoptotic pathways and protects against C2-ceramide-induced Cath.a differentiated (CAD) cell death via the PI3K/AKT pathway (Sanchez-Mora et al., 2012).

3.2.3. Interaction between the PI3K/AKT signaling pathway and DJ-1

In addition to PINK1 and LRRK2, DJ-1 has also been proposed to be involved in the PI3K/AKT pathway. DJ-1 was first identified as a negative regulator of the tumor suppressor, PTEN, and as a positive regulator of AKT (Sitaram et al., 2009). Yang et al. (2005) demonstrated that inactivation of DJ-1 leads to impairments of PI3K/AKT signaling and the oxidative stress response in mammalian systems (Yang et al., 2005). Recently, Aleyasin and colleagues observed that DJ-1 promotes AKT phosphorylation in response to oxidative stress-induced by H₂O₂ both *in vitro* and *in vivo* following MPTP treatment. This suggests that DJ-1 is crucial for full activation of AKT upon oxidative injury and it is speculated that

this partially underlies the neuroprotection of DJ-1 (Aleyasin et al., 2010). However, this remains to be confirmed in other cellular and animal models.

3.2.4. Interaction between the PI3K/AKT and JNK signaling pathways

The PI3K/AKT and JNK pathways are the two most important signaling cascades involved in PD, and cross-talk between these two pathways determines the equilibrium between neuronal death and survival. AKT negatively regulates the phosphorylation and activation of JNK by directly phosphorylating apoptosis signal-regulating kinase 1 (ASK1) and JNK-interacting protein 1 (JIP1), thus preventing cell apoptosis and promoting cell survival (Kim et al., 2001, 2002). Interestingly, in 1-methyl-4-phenylpyridinium (MPP⁺)-induced human neuroblastoma SH-EP1 cells, IGF-1 potentiated PI3K/AKT activity negatively regulates the JNK-related apoptotic pathway. Furthermore, this negative regulation has been shown to be mediated by AKT-dependent GSK3 β inactivation (Wang et al., 2010). Thus, the balance between activation of the AKT pathway and inhibition of the JNK pathway could directly determine cell fate (death or survival) in Parkinsonism settings.

4. Summary

Remarkable progress and advances have been made recently in elucidation of the mechanism of PD pathogenesis, particularly with the aid of extended knowledge of PD-linked genes. These involve impairments of UPS and ALP, mitochondrial metabolism disturbance and altered kinase activity. Evidence for the role of disturbance of kinase activity in PD is mounting. Increasingly, research is being focused on several familial mutations in genes encoding kinases (PINK1, LRRK2, GAK) and the JNK/ERK/p38/AKT signaling pathways. Through elucidation of the physiological functions of protein kinases (PINK1, LRRK2 and GAK) and associated kinase signaling cascades (JNK/ERK/p38 and AKT pathways) and the pathological mechanisms triggered by the mutated counterparts, it will be possible to establish, not only the contribution of kinase dysregulation to the etiology of PD, but also to develop kinase inhibitors as a therapeutic strategy for the

Table 2

Novel neuroprotective agents for intervention in kinases and kinase signaling pathways.

Categories	Compound	Injury model	Possible mechanisms
LRRK2 inhibitor	Selected compound 4	The cell-based assay via overexpression of LRRK2 mutant, G2019S	Hydrogen bond interactions between compound 4 and Ala 1950 residue in the backbone of the ATP binding pocket of LRRK2 kinase domain (Yun et al., 2011)
	CZC-25146	Mutant LRRK2-induced injury of cultured rodent and human neurons	Selective LRRK2 inhibition can attenuate mutant LRRK2-dependent toxicity in human neurons (Ramsden et al., 2011)
	Sorafenib and GW5074	G2019S induced neurodegeneration in <i>Caenorhabditis elegans</i> and in <i>Drosophila</i> MPTP-induced mice	Increasing kinase activity of LRRK2 is neurotoxic and inhibition of LRRK2 activity has a disease-modifying effect (Liu et al., 2011)
JNK inhibitor	SP600125		Reduced the levels of c-Jun phosphorylation blocked p-c-Jun translocation from the cytoplasm to the nucleus in dopaminergic neurons of substantia nigra, mitigated the loss of dopaminergic neurons and improved motor function (Wang et al., 2009)
	Tat-JBD	MPTP-induced mice	Perturbation of the assembly of the JIP-1-JNKs complex, inhibits activation of JNKs induced by MPTP and consequently diminishes phosphorylation of c-Jun. Furthermore, it inhibits phosphorylation of Bcl-2 and release of Bax from Bcl-2/Bax dimers, sequentially attenuates translocation of Bax to mitochondria, release of cytochrome c, activation of caspase3 and the hydrolyzation of poly-ADP-ribose-polymerase (Pan et al., 2010)
ERK inhibitor	SL327	6-OHDA-lesioned mice	Inactivation of ERK1/2 during chronic l-DOPA treatment counteracts the induction dyskinesia (Santini et al., 2007)
P38 inhibitor	SB239063	MPP ⁺ -induced primary dopaminergic neurons	Prevention of the downstream phosphorylation of p53 and its translocation to the nucleus <i>in vivo</i> , in the ventral midbrain, and inhibition of apoptosis progress (Karunakaran et al., 2008)
PI3K activator	TSG	MPP ⁺ -induced PC12 cells	Attenuation of loss of cell viability and release of lactate dehydrogenase (LDH), and inhibition of apoptosis in a dose-dependent manner (Qin et al., 2011)

treatment of PD (Cuny, 2009). Serial disease-modifying agents, including kinase modulators agents (LRRK2, JNK, p38 and ERK inhibitors and PI3K activators) (Table 2) are promising therapeutic effects on the motor symptoms and LID complications of PD in future. Meanwhile, phosphatases that reduce phosphorylation are equally important in maintaining phosphorylation status as the kinases. The balance of phosphatase and kinase activities is altered in PD and phosphorylation status of key protein (α -synuclein, for example) changed, indicating post-translational modification in the disease pathogenesis (Braithwaite et al., 2012a,b). Therefore, modulating phosphatases has the same potential as kinases for the next generation of therapeutic targets to cure PD. In conclusion, studies on kinase signaling pathways not only improve our understanding of pathogenesis of PD, but also provide candidates for therapeutic intervention in patients with PD.

Conflict of interest statement

There is no conflict of interest.

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